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SECTION: G

PROBLEM STATEMENT #10

Campus & Academic Operations

(Student Club Event Ticketing & Budget Portal)

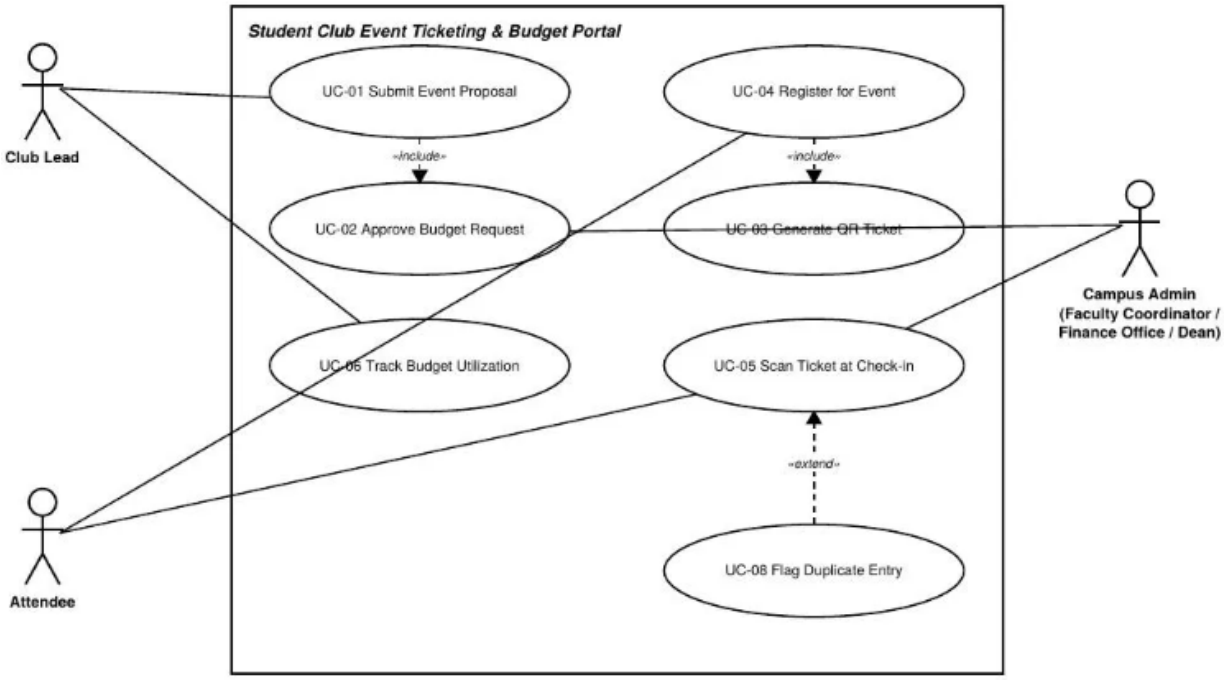
DELIVERABLES:

1. REQUIREMENTS TABLE:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall route event budget proposals through a multi-stage approval workflow (Faculty Coordinator → Finance Office → Dean).	High	Pass: Status transitions sequentially upon digital sign-off. Fail: Request reaches Dean without Finance Office sign-off.	Ensures financial accountability and prevents unauthorized budget approval.	Clear and testable. Could add a time limit for each approval stage.
FR-002	Functional	The system shall allow a Club Lead to submit a new event proposal containing event name, date, venue, expected attendance, and requested budget.	High	Pass: Proposal is saved with a unique Proposal ID and status "Submitted" visible on the Club Lead's dashboard. Fail: Proposal is accepted despite a missing mandatory field.	Establishes the entry point for the entire budget and ticketing workflow.	Good. List the required fields in the description, not just the acceptance criteria.
FR-003	Functional	The system shall automatically generate a unique, non-forgeable QR-code ticket for each attendee who registers for an approved event.	High	Pass: Each registration produces one QR code containing a unique ticket ID, event ID, and cryptographic signature. Fail: Two attendees receive identical codes, or a code is generated for an unapproved event.	Enables secure, verifiable entry management at events.	"Non-forgeable" is a bit unclear.
FR-004	Functional	The system shall allow an Attendee to register for any event with "Approved" status and receive their ticket by email within 5 minutes of registration.	Medium	Pass: Confirmation email with QR ticket attached arrives within 5 minutes of successful registration. Fail: No email is received, or registration succeeds for a non-approved event.	Provides attendees a reliable, self-service way to obtain event access.	What happens if the email fails to send? Might need a retry option.
FR-005	Functional	The system shall validate each QR ticket at venue check-in and reject a ticket that has already been scanned or does not match the event.	High	Pass: A duplicate or mismatched scan is rejected with an on-screen "Access Denied" message and logged. Fail: A duplicate ticket is accepted, permitting a second entry.	Prevents ticket fraud and duplication; enforces one-entry-per-ticket policy.	Testable.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
NFR-001	Nonfunctional	The QR-code ticket scanner API shall validate entry authenticity in under 100 ms at venue check-in points.	High	Pass: Benchmark tests confirm <100 ms latency and security standards under simulated peak load. Fail: Validation exceeds 100 ms, or peak-load testing reveals a security bypass.	Ensures fast, secure entry during high-traffic check-in windows without bottlenecking queues.	Good measurable NFR.
NFR-002	Nonfunctional	The system shall cryptographically sign every QR ticket so that any forged, altered, or duplicated code is rejected by the scanner.	High	Pass: A penetration test using 50 tampered/forged QR codes results in 0 codes accepted. Fail: Any forged or tampered code is accepted as valid.	Protects ticketing integrity and prevents unauthorized event access via forged tickets.	Solid requirement.

2. UML USE – CASE DIAGRAM:



3. USE- CASE FLOW SPECIFICATION

Actors

Primary: Attendee (presents ticket). Secondary: Campus Admin / Gate Staff (operates scanner).

Preconditions

1. The Attendee has successfully registered for the event and holds a valid QR-code ticket (per UC-04, which includes UC-03 – Generate QR Ticket).
2. The event's budget proposal has status “Approved” and the event is scheduled for the current date.
3. The venue check-in scanner is powered on and connected to the ticketing system.

Postconditions

1. On success: the ticket status is updated to “Used” with a timestamp, and the attendee is granted entry.
2. On failure: the attempt is logged, the ticket status is unchanged, and entry is denied.

Main Success Scenario

1. Attendee presents their QR ticket (digital or printed) to the scanner at the venue entrance.
2. System captures and decodes the QR-code image.
3. System verifies the code's cryptographic signature and confirms it matches the current event (NFR-002).
4. System checks the ticket's status in the database and confirms it is “Unused.”
5. System returns a validation result to the scanner in under 100 ms (NFR-001).
6. System marks the ticket as “Used” and records the entry timestamp.
7. Gate Staff sees a green “Access Granted” confirmation and allows the Attendee to enter. Use case ends successfully.

Alternate Flow – 4a: Duplicate or Invalid Ticket

- 4a1. System detects the ticket status as “Used”, or the signature/event as invalid.
- 4a2. System displays a red “Access Denied – Duplicate/Invalid Ticket” alert to Gate Staff and logs the attempt (UC-08).
- 4a3. Gate Staff manually checks the Attendee's ID against the registration list.
- 4a4. If the mismatch is confirmed as an honest error (e.g., the attendee's own ticket was scanned twice by accident), Gate Staff may issue a one-time manual override, logged with staff ID. Otherwise, entry is denied and the incident is flagged for review. Use case ends.